

PART-B

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	3	7	5	0	15

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yx} \\ S_{yx} & S_{xx} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

Given $S = \begin{pmatrix} S_{yy} & S_{xy} \\ S_{yx} & S_{xx} \end{pmatrix}$ $S_{xy} = (S_{yx})^T$

Now $\sigma_1^2 = \begin{pmatrix} S_{yy}^{-\frac{1}{2}} & S_{yx} & S_{xx}^{-1} & S_{xy} S_{yy}^{-\frac{1}{2}} \end{pmatrix}$ Since this will be a matrix of

size 1×1 hence for a matrix of size 1×1 its value itself is equal to its eigenvalue.

to get $S_{yy} = \hat{\sigma}^2$

$\therefore \sigma_1^2 = \frac{1}{\hat{\sigma}^2} (S_{yx} S_{xx}^{-1} S_{xy})$

So $R^2 = 1 - \frac{S_{yy} - S_{yx} S_{xx}^{-1} S_{xy}}{S_{yy}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

Now to prove

03 $\sigma_1^2 = \frac{1}{\hat{\sigma}^2} (S_{yx} S_{xx}^{-1} S_{xy}) = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

Now $\hat{y} = Hy$ where $H = \frac{X(X^T X)^{-1} X^T}{\text{based on MLR}}$ (here X is the matrix)

$\therefore y - \hat{y} = y(I - H)y$

$\therefore \sum (y_i - \hat{y}_i)^2 = \sum y_i^2 + \hat{y}_i^2 - 2y_i \hat{y}_i$
 $= \sum y_i^2 - \hat{y}_i (y_i - 2y_i)$

$$= \sum y_i^2 - \hat{y}_i (-e_i - y_i)$$

$$= \sum y_i^2 - \hat{y}_i y_i \quad \therefore \sum \hat{y}_i e_i = 0$$

$$= \sum y_i (y_i - \hat{y}_i)$$

$$= \sum y_i e_i$$

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix S . The eigenvalues of S are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are e_1, e_2, e_3 , and e_4 . You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{L}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{L}$ be $\tilde{l}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of S .

Q1) $\tilde{L} = [\sqrt{\lambda_3} e_3 \mid \sqrt{\lambda_2} e_2]$

$\tilde{L} = [\sqrt{4.2} e_3 \mid \sqrt{2.1} e_2]$

Thus $\tilde{L} = [2.0493 e_3 \mid 1.4491 e_2]$

is the final loading matrix

(Here we considered two largest eigenvalues of S)

b) First row is $[0.8, -0.4]$

Now $\tilde{h}_1^2 = d_{11}^2 + d_{12}^2 + \dots + d_{1m}^2$

~~$\tilde{h}_1^2 =$~~
 $\therefore \tilde{h}_1^2 = d_{11}^2 + d_{12}^2 = (0.8)^2 + (-0.4)^2$

$= 0.64 + 0.16$

$= \boxed{0.8}$

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$\therefore \boxed{\tilde{h}_1^2 = 0.8}$

c) Now $\tilde{L} = [\sqrt{\lambda_1} \hat{e}_1 \mid \sqrt{\lambda_2} \hat{e}_2]$ where λ_1, λ_2 are two largest eigenvalues of S and $\hat{e}_1$ and $\hat{e}_2$ are their corresponding eigenvectors.

(Here we have assumed $(\lambda_1, e_1) = (4.2, e_3)$
 and $(\lambda_2, e_2) = (2.1, e_2)$)

Now $\tilde{h}_i^2 = \sum_{j=1}^m d_{ij}^2$

$\therefore \sum_{i=1}^p \tilde{h}_i^2 = \sum_{i=1}^p \sum_{j=1}^m d_{ij}^2 = \text{i.e. } \sum_{i=1}^p \tilde{h}_i^2 \text{ is simply the sum of square of all elements of the } L \text{ matrix}$

$\therefore$ Our $\tilde{U}$ matrix looks like

$$\begin{bmatrix} \sqrt{\lambda_1} e_{11} & \sqrt{\lambda_2} e_{21} \\ \sqrt{\lambda_1} e_{12} & \sqrt{\lambda_2} e_{22} \\ \sqrt{\lambda_1} e_{13} & \sqrt{\lambda_2} e_{23} \\ \sqrt{\lambda_1} e_{14} & \sqrt{\lambda_2} e_{24} \end{bmatrix}$$

$\therefore$ ~~Sum of squares~~ $\therefore \sum_{i=1}^p \sum_{j=1}^m d_{ij}^2 = \lambda_1 (e_{11}^2 + e_{12}^2 + e_{13}^2 + e_{14}^2) + \lambda_2 (e_{21}^2 + e_{22}^2 + e_{23}^2 + e_{24}^2)$

But since the eigenvectors are normalized i.e.

$$\|e_i\| = 1 \quad \forall i = 1, 2, 3, 4 \quad \text{Hence}$$

$$e_{11}^2 + e_{12}^2 + e_{13}^2 + e_{14}^2 = 1 \quad \text{and}$$

$$e_{21}^2 + e_{22}^2 + e_{23}^2 + e_{24}^2 = 1$$

$$\therefore \sum_{i=1}^p \sum_{j=1}^m d_{ij}^2 = \lambda_1 (1) + \lambda_2 (1) = \lambda_1 + \lambda_2$$

$$\therefore \sum_{i=1}^p \tilde{h}_i^2 = \sum_{j=1}^p \sum_{i=1}^m d_{ij}^2 = \lambda_1 + \lambda_2$$

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$$\sum_{i=1}^p \tilde{h}_i^2 = \lambda_1 + \lambda_2 = \text{Sum of two largest eigenvalues of } S$$

Hence proved.

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

Now we know that

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \mu^2)$$

$$\Rightarrow \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \mu^2)$$

By and $\therefore g(x_n) \approx g(\mu) \approx g'(\mu)(x_n - \mu)$

By Univariate Delta Theorem,

$$\sqrt{n} \sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N(0, (g'(\mu))^2 \mu^2)$$

$$\therefore \text{Asymptotic Variance of } g(\bar{X}_n) = \frac{(g'(\mu))^2 \mu^2}{n}$$

$\therefore$ For the asymptotic variance to not depend on μ .

$$\Rightarrow (g'(\mu))^2 \mu^2 = c \quad (\text{A fixed constant})$$

$$\therefore g'(\mu) = \frac{c_1}{\mu} \quad c_1 = \sqrt{c}$$

$$\therefore \frac{dg(\mu)}{d\mu} = \frac{c_1}{\mu}$$

$$\therefore dg(\mu) = \frac{c_1 d\mu}{\mu} \Rightarrow \left(\int dg(\mu) = \int \frac{c_1 d\mu}{\mu} \right)$$

$$\therefore dg(\mu) \therefore g(\mu) = C \ln(\mu) + D \quad (\text{On integration})$$

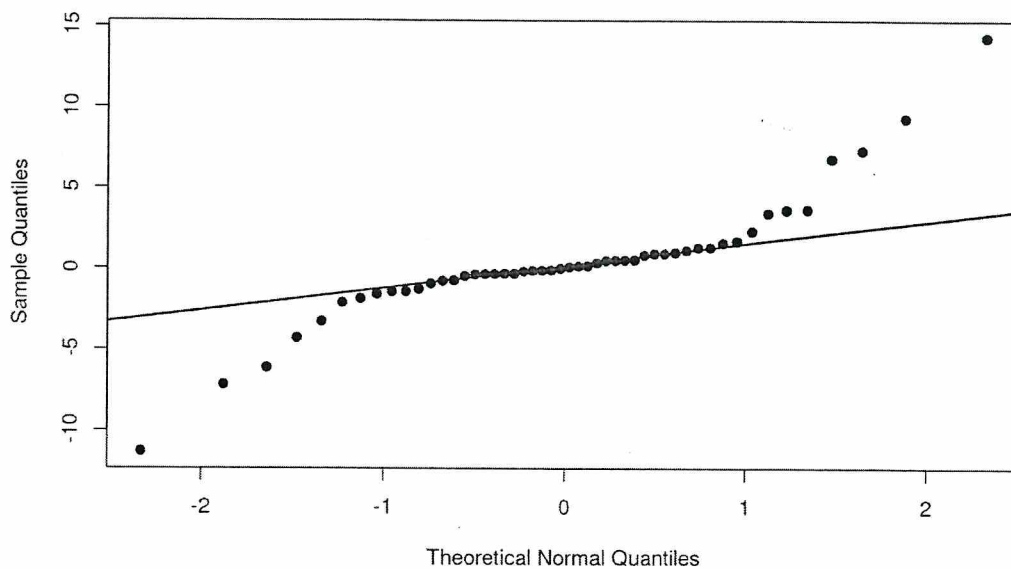
For convenience we can take, $C=1$ and $D=0$

$$\therefore g(\mu) = \ln(\mu)$$

Thus the function $g(\bar{X}_n) = \ln(\bar{X}_n)$ is the required variance stabilizing function.

05

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



This is an example of a short tailed distribution in which very few values lie on the outskirts. This is a manageable and the normality assumption won't produce large errors.

If it's normal then Ideal

If it's lognormal i.e. skewed then we can do a transformation of the response

If it's long tailed we need to change the mathematical model.

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ADDITIONAL PAGES FOR ANSWERS
